

Ex. 7.1

Consider the daily returns of GE stock from January 2, 1998, to December 31, 2008. The data can be obtained from CRSP or the file `d-ge9808.txt`. Convert the simple returns into log returns. Suppose that you hold a long position on the stock valued at \$1 million. Use the tail probability 0.01. Compute the value at risk of your position for 1-day horizon and 15-day horizon using the following methods:

- The RiskMetrics method.
- A Gaussian ARMA–GARCH model.
- An ARMA–GARCH model with a Student- t distribution. You should also estimate the degrees of freedom.
- The traditional extreme value theory with subperiod length $n = 21$

As customary, we begin by setting the working directory in order to ensure a consistent and reproducible execution environment for the analysis.

Subsequently, all the required R packages are installed (if not already available in the local library) and loaded into the session.

Following the data specifications of Assignment 2, we load the daily price series and transform it into continuously compounded returns.

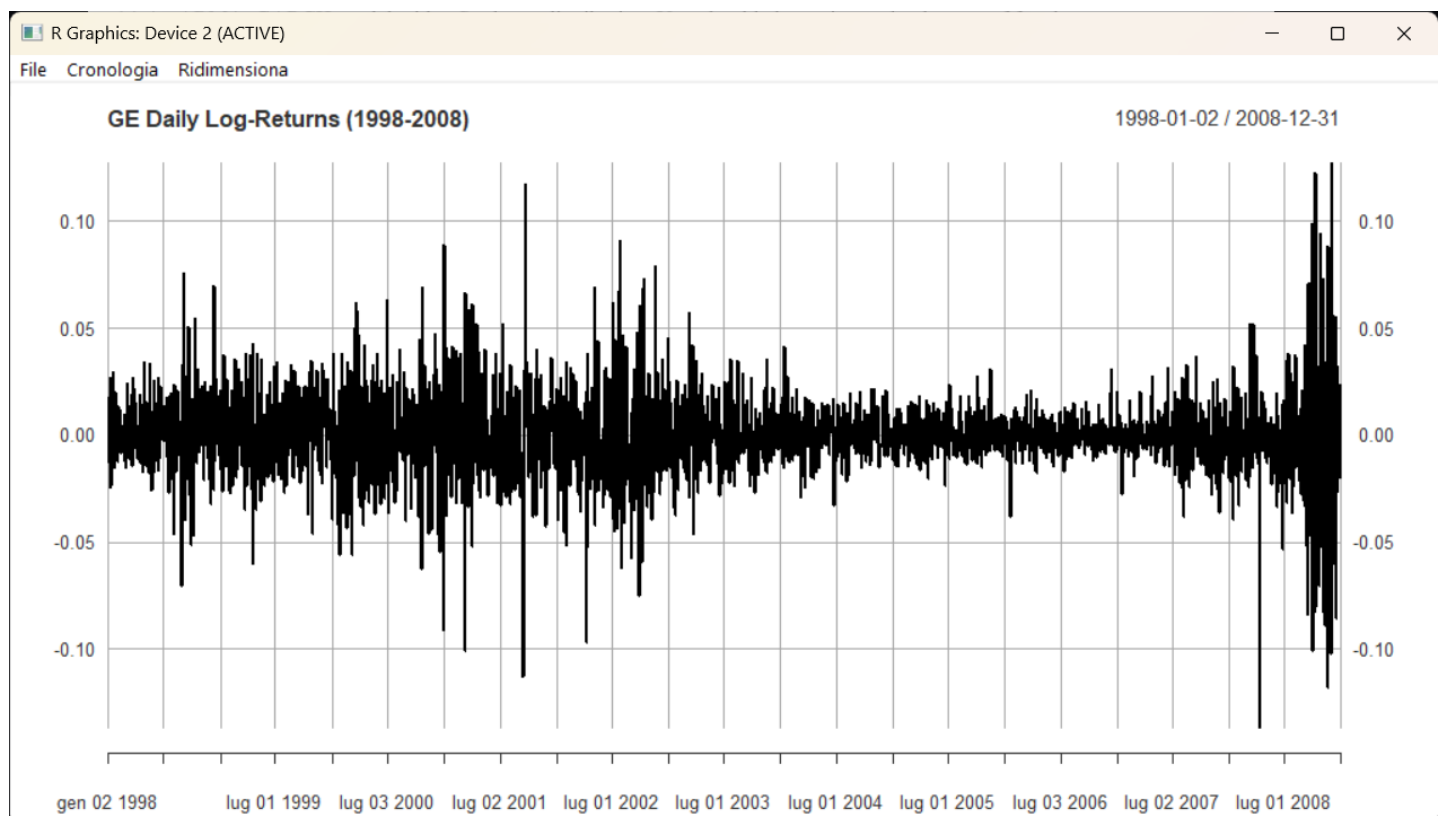
```
14 ge_raw <- read.table("d-ge9808.txt", header = TRUE)
15 ge_raw$date <- as.Date(as.character(ge_raw$date), format = "%Y%m%d")
16 ge_ret <- log(1 + ge_raw$ge)
17 ge_ts <- xts(ge_ret, order.by = ge_raw$date)
18 colnames(ge_ts) <- "GE_LogRet"
```

The resulting return series is then converted into an xts object, which allows for a proper time-series representation indexed by calendar dates and is required for the subsequent implementation of volatility models and risk measures.

Finally, we define the key investment and risk parameters used throughout the assignment, namely the portfolio position size and the confidence level associated with tail risk measures, in accordance with the Value-at-Risk framework discussed in the reference textbooks.

```
20 # Investment and risk Parameters
21 Size <- 1000000 # Position size: $1,000,000
22 alpha <- 0.01 # Significance level: 1%

24 # Plot the returns series
25 windows()
26 plot(ge_ts, main = "GE Daily Log>Returns (1998-2008)")
```



The daily log-return series is plotted over the full sample period in order to visually inspect its time-series behavior. The plot shows the daily log-returns of GE over the period 1998–2008. The series fluctuates around zero and displays periods of higher and lower variability. Large return realizations are observed at several points in the sample, with particularly pronounced movements toward the end of the period.

```

28 ##
29 #The RiskMetrics method
30 spec_rm <- ugarchspec(
31   mean.model = list(armaOrder = c(0,0), include.mean = FALSE),
32   variance.model = list(model = "iGARCH", garchOrder = c(1,1)),
33   fixed.pars = list(omega = 0))
34
35 # Estimate the model
36 rm <- ugarchfit(spec = spec_rm, data = ge_ts)
37 rm

```

```

*-----*
*          GARCH Model Fit          *
*-----*

Conditional Variance Dynamics
-----
GARCH Model      : iGARCH(1,1)
Mean Model       : ARFIMA(0,0,0)
Distribution      : norm

Optimal Parameters
-----
      Estimate Std. Error t value Pr(>|t|)
omega    0.000000      NA      NA      NA
alpha1   0.039603    0.004946   8.0073    0
beta1    0.960397      NA      NA      NA

Robust Standard Errors:
      Estimate Std. Error t value Pr(>|t|)
omega    0.000000      NA      NA      NA
alpha1   0.039603    0.009048   4.3771 1.2e-05
beta1    0.960397      NA      NA      NA

LogLikelihood : 7484.091

```

The RiskMetrics specification is defined by imposing a zero-mean process and an iGARCH(1,1) structure for the conditional variance, with the constant term fixed to zero. The model is then estimated using the full sample of GE daily log-returns.

The conditional variance is modeled using an iGARCH(1,1) specification with normally distributed innovations and no dynamics in the conditional mean. The estimated parameters satisfy the unit-root condition in the variance equation, with $\alpha_1 + \beta_1 = 1$, which is consistent with the RiskMetrics EWMA formulation.

```

39 # Forecast volatility for the next day (t+1)
40 fc_rm <- ugarchforecast(rm, n.ahead = 1)
41 sigma_rm_1d <- sigma(fc_rm)[1]
42
43 # VaR Calculation , remember that VaR = Position Size * Sigma * Quantile
44 z_score <- qnorm(1 - alpha)
45 VaR_rm_1d <- Size * sigma_rm_1d * z_score
46 VaR_rm_1d
47
48 # 15-day VaR
49 VaR_rm_15d <- VaR_rm_1d * sqrt(15)
50 VaR_rm_15d

> VaR_rm_1d
[1] 115727.5
> VaR_rm_15d
[1] 448210.7

```

The conditional volatility is forecast one step ahead in order to obtain a forward-looking measure of risk. Since the Value-at-Risk is defined as a prospective risk measure, it must be based on the predicted conditional volatility for the next period rather than on the in-sample estimate.

Under the normality assumption of the RiskMetrics framework, the one-day Value-at-Risk is computed by scaling the one-step-ahead volatility forecast by the appropriate standard normal quantile and the portfolio position size.

Finally, the multi-day Value-at-Risk is obtained by applying the square-root-of-time rule to the one-day measure.

The one-day Value-at-Risk at the 1% confidence level amounts to approximately 115,728, representing the maximum expected loss over the next trading day under the RiskMetrics assumptions. Applying the square-root-of-time rule, the corresponding 15-day Value-at-Risk increases to approximately 448,211, reflecting the accumulation of risk over a longer holding period.

```
53 #Gaussian ARMA-GARCH Model
54 spec_gauss <- ugarchspec(
55   mean.model = list(armaOrder = c(1, 0), include.mean = TRUE),
56   variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
57   distribution.model = "norm")
58
59 # Estimate the model
60 gauss <- ugarchfit(spec = spec_gauss, data = ge_ts)
61 gauss
62
63 # Forecast the next 15 days
64 fc_gauss <- ugarchforecast(fit_gauss, n.ahead = 15)
65 mu_pred_gauss <- fitted(fc_gauss) # Vector of predicted means
66 sigma_pred_gauss <- sigma(fc_gauss) # Vector of predicted sigmas
```

```
*-----*
*          GARCH Model Fit          *
*-----*
```

Conditional Variance Dynamics

```
GARCH Model      : sGARCH(1,1)
Mean Model       : ARFIMA(1,0,0)
Distribution      : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(> t)
mu	0.000389	0.000243	1.60346	0.108832
ar1	-0.011450	0.019872	-0.57620	0.564477
omega	0.000001	0.000001	0.60755	0.543483
alpha1	0.050761	0.011161	4.54797	0.000005
beta1	0.948239	0.011686	81.14640	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(> t)
mu	0.000389	0.000272	1.428414	0.15317
ar1	-0.011450	0.019388	-0.590588	0.55480
omega	0.000001	0.000010	0.078605	0.93735
alpha1	0.050761	0.096761	0.524606	0.59986
beta1	0.948239	0.099784	9.502881	0.00000

LogLikelihood : 7492.921

A Gaussian ARMA–GARCH model is specified by allowing for an AR(1) component in the conditional mean and a standard GARCH(1,1) structure for the conditional variance.

The model is estimated on the full return series and subsequently used to generate multi-step-ahead forecasts of both the conditional mean and the conditional volatility over a 15-day horizon.

The estimated variance parameters indicate a high degree of persistence in volatility dynamics, with the sum of the GARCH coefficients close to unity.

```
68 #1-Day VaR
69 z_alpha <- qnorm(alpha)
70 VaR_gauss_1d <- -Size * (as.numeric(mu_pred_gauss[1]) + as.numeric(sigma_pred_gauss[1]) * z_alpha)
71 VaR_gauss_1d
72
73 #15-Day VaR
74 mu_15_gauss <- sum(mu_pred_gauss)
75
76 # Volatility: Square root of sum of variances (assuming independence over time)
77 sigma_15_gauss <- sqrt(sum(sigma_pred_gauss^2))
78 VaR_gauss_15d <- -Size * (mu_15_gauss + sigma_15_gauss * z_alpha)
79 VaR_gauss_15d
```

```
> VaR_gauss_1d
[1] 111456.6
> VaR_gauss_15d
[1] 425600.8
```

The one-day Value-at-Risk is computed using the Gaussian ARMA–GARCH forecasts of the conditional mean and volatility. For the multi-day horizon, the forecasted conditional means are aggregated linearly, while the conditional variances are aggregated under the assumption of temporal independence. The 15-day Value-at-Risk is then obtained by combining the aggregated mean and volatility forecasts at the chosen confidence level.

Under the Gaussian ARMA–GARCH specification, the one-day Value-at-Risk at the 1% confidence level is approximately 111,457. Over a 15-day horizon, the Value-at-Risk increases to approximately 425,601, reflecting the accumulation of risk over time based on the model's forecasted dynamics.

```

81 ##
82 #Student-t ARMA-GARCH Model
83 spec_t <- ugarchspec(
84   mean.model = list(armaOrder = c(1, 0), include.mean = TRUE),
85   variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
86   distribution.model = "std")
87
88 fit_t <- ugarchfit(spec = spec_t, data = ge_ts)
89
90 # Extract estimated Degrees of Freedom (Shape parameter)
91 nu_est <- coef(fit_t)["shape"]
92
93 # Forecast for 15 days
94 fc_t <- ugarchforecast(fit_t, n.ahead = 15)
95 mu_pred_t <- fitted(fc_t)
96 sigma_pred_t <- sigma(fc_t)
97
98 # Calculate appropriate quantile using the estimated Degrees of Freedom
99 t_alpha_quantile <- qdist("std", p = alpha, mu = 0, sigma = 1, shape = nu_est)

```

A Student-t ARMA–GARCH model is specified by allowing for an AR(1) component in the conditional mean and a GARCH(1,1) structure for the conditional variance, while relaxing the normality assumption for the innovations.

The model is estimated on the full return series, and the degrees of freedom of the Student-t distribution are extracted from the fitted parameters.

Multi-step-ahead forecasts of the conditional mean and volatility are then generated over a 15-day horizon. The Value-at-Risk quantile is computed using the Student-t distribution with the estimated degrees of freedom, ensuring consistency between the assumed return distribution and the tail risk measure.

```

101 # --- 1-Day VaR ---
102 VaR_t_1d <- -Size * (as.numeric(mu_pred_t[1]) + as.numeric(sigma_pred_t[1]) * t_alpha_quantil
103 VaR_t_1d
104
105 # --- 15-Day VaR ---
106 mu_15_t <- sum(mu_pred_t)
107 sigma_15_t <- sqrt(sum(sigma_pred_t^2))
108 VaR_t_15d <- -Size * (mu_15_t + sigma_15_t * t_alpha_quantile)
109 VaR_t_15d

```

```

> VaR_t_1d
[1] 121199.8
> VaR_t_15d
[1] 463887.2

```

Under the Student-t ARMA–GARCH specification, the one-day Value-at-Risk at the 1% confidence level is approximately 121,200. Over a 15-day horizon, the Value-at-Risk increases to approximately 463,887, reflecting the heavier tails implied by the Student-t innovation distribution.

```

111 ##
112 #Traditional EVT with subperiod n=21
113 losses <- -as.numeric(ge_ts)
114 block_size <- 21
115
116 # Fit GEV distribution to block maxima
117 fit_gev <- gev(losses, block = block_size)
118
119 # Extract GEV parameters (Xi, Sigma, Mu)
120 xi_hat <- fit_gev$par.ests["xi"]
121 sigma_hat <- fit_gev$par.ests["sigma"]
122 mu_hat <- fit_gev$par.ests["mu"]
123
124 #VaR
125 p_star <- (1 - alpha)^block_size

```

An Extreme Value Theory approach is implemented using a block maxima method with non-overlapping blocks of 21 observations, corresponding to approximately one trading month.

The Generalized Extreme Value (GEV) distribution is fitted to the block maxima using the evir package, and the estimated shape, scale, and location parameters are extracted from the fitted model.

```

127 # Calculate quantile manually using the inverse GEV formula:
128 if(abs(xi_hat) > 1e-6) {
129   VaR_evt_val <- mu_hat - (sigma_hat/xi_hat) * (1 - (-log(p_star))^(xi_hat))
130 } else {
131   # Gumbel case (limit as xi -> 0)
132   VaR_evt_val <- mu_hat - sigma_hat * log(-log(p_star))
133 }
134
135 # Convert to monetary value
136 VaR_evt_1d <- Size * VaR_evt_val
137 VaR_evt_15d <- VaR_evt_1d * sqrt(15)
> VaR_evt_1d
      mu
46439.82
> VaR_evt_15d
      mu
179860.6

```

The Value-at-Risk is then computed by evaluating the inverse GEV quantile function at the appropriate probability level, adjusted for the block size.

The computation accounts for both the general case and the limiting Gumbel case when the shape parameter approaches zero.

The resulting quantile is scaled by the portfolio size to obtain a one-day monetary Value-at-Risk. Finally, the 15-day Value-at-Risk is obtained by applying the square-root-of-time rule.

Under the EVT block maxima approach, the one-day Value-at-Risk at the 1% confidence level is approximately 46,440.

Applying the square-root-of-time rule, the corresponding 15-day Value-at-Risk increases to approximately 179,861.

Ex. 10.1

Consider the monthly simple returns, including dividends, of IBM stock, Hewlett-Packard (HPQ) stock, and the S&P composite index from January 1962 to December 2008 for 564 observations. The returns are in the file `m-ibmhpqsp6208.txt`. Transform into log returns in percentages. Use the exponentially weighted moving-average method to obtain a multivariate volatility series for the three return series. What is the estimated λ ? Plot the three volatility series.

```
144 # Ex 10.1
145
146 data <- read.table("m-ibmhpqsp6208.txt", header = TRUE)
147 returns_simple <- data[, c("IBM", "HPQ", "SP")]
148 returns_log <- 100 * log(1 + returns_simple)
149
150 # Remove missing values
151 returns_log <- na.omit(returns_log)
152
153 # Center returns (zero mean)
154 returns_centered <- scale(returns_log, center = TRUE, scale = FALSE)
```

The data table containing the monthly simple returns of IBM, HPQ, and the S&P composite index is loaded into the workspace.

The returns are then transformed into log-returns expressed in percentages.

Missing observations are removed, and the return series are centered by subtracting their sample means in order to obtain zero-mean returns for the subsequent multivariate EWMA estimation.

```
156 #EWMA negative log-likelihood
157 ewma_loglik <- function(lambda, data) {
158   Sigma <- cov(data)
159   ll <- 0
160
161   for (t in 2:nrow(data)) {
162     r <- as.numeric(data[t - 1, ])
163
164     # Penalize non-positive definite Sigma
165     if (det(Sigma) <= 0) return(1e6)
166
167     ll <- ll +
168       log(det(Sigma)) +
169       t(r) %*% solve(Sigma) %*% r
170
171     Sigma <- (1 - lambda) * (r %*% t(r)) + lambda * Sigma
172   }
173   return(0.5 * ll)
174 }
175
176 }
```

A negative log-likelihood function for the multivariate EWMA model is defined.

The conditional covariance matrix is initialized using the sample covariance of the centered returns and updated recursively according to the EWMA scheme.

At each time step, the contribution to the log-likelihood is computed using the multivariate normal density, and non-positive definite covariance matrices are penalized to ensure numerical stability.

The function returns the scaled negative log-likelihood value, which is subsequently minimized with respect to the smoothing parameter λ .

```
178 # Estimation of lambda
179
180 opt <- optimize(
181   ewma_loglik,
182   interval = c(0.85, 0.98),
183   data = returns_centered)
184
185 lambda_est <- opt$minimum
186 lambda_est
```

The smoothing parameter λ of the multivariate EWMA model is estimated by minimizing the negative log-likelihood function over a restricted interval.

The optimization is performed using a one-dimensional numerical search, and the value of λ that minimizes the objective function is retained as the estimated smoothing parameter.

The estimated value of the smoothing parameter is $\lambda = 0.9691$.

This value indicates a high degree of persistence in the conditional covariance dynamics, implying that past observations receive substantial weight in the EWMA updating scheme.

Consequently, the estimated volatility responds smoothly to new return realizations and exhibits slow decay over time.

```
> lambda_est  
[1] 0.9691408
```

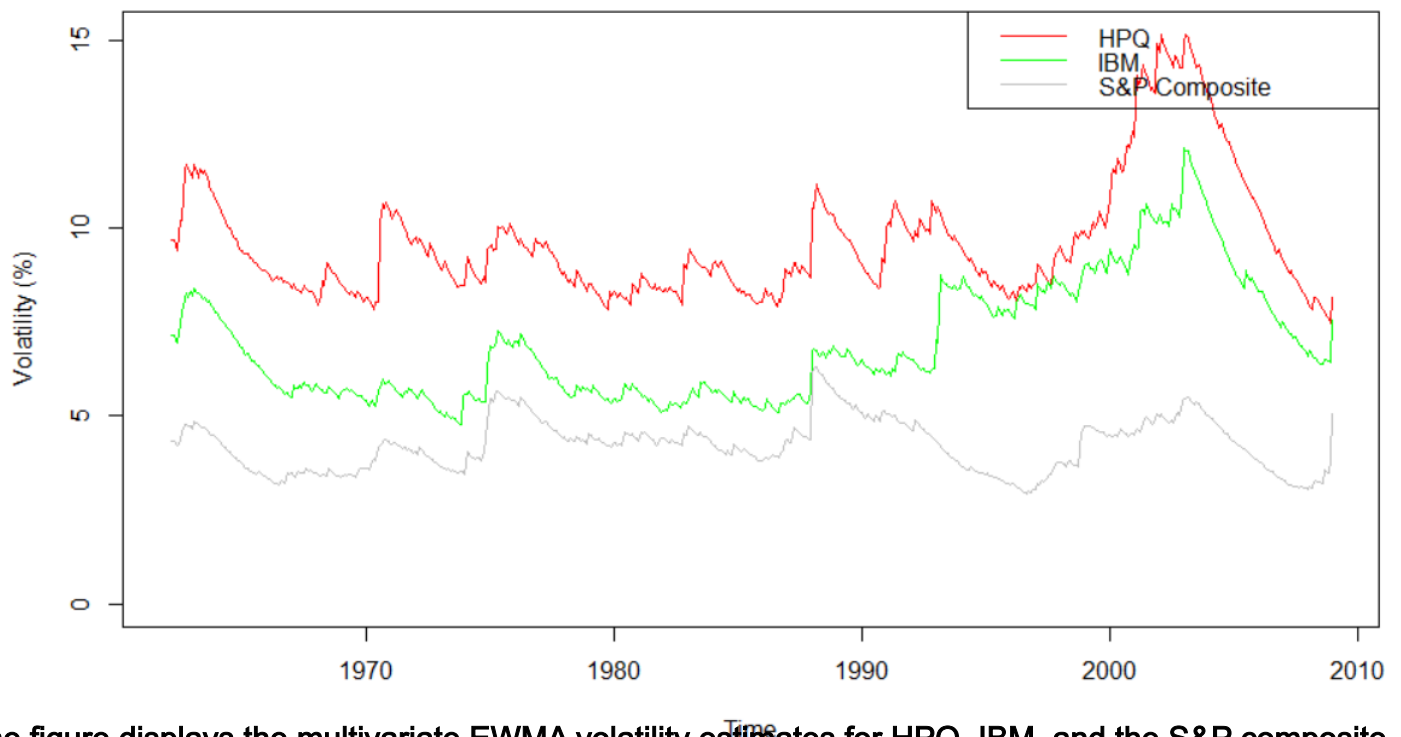
```
188 #EWMA Volatility estimation  
189 volatilities <- matrix(NA, nrow = T, ncol = ncol(returns_centered))  
190 colnames(volatilities) <- colnames(returns_log)  
191  
192 Sigma <- cov(returns_centered)  
193  
194 for (t in 2:T) {  
195  
196   volatilities[t, ] <- sqrt(diag(Sigma))  
197  
198   r <- as.numeric(returns_centered[t - 1, ])  
199   Sigma <- (1 - lambda_est) * (r %*% t(r)) + lambda_est * Sigma  
200 }
```

The multivariate EWMA volatility series are computed by recursively updating the conditional covariance matrix using the estimated smoothing parameter λ .

At each time step, the marginal volatilities are obtained as the square roots of the diagonal elements of the conditional covariance matrix.

The resulting volatility series are then plotted for the three assets over the sample period.

```
202 # Plot  
203 dates <- as.Date(as.character(data$date), format = "%Y%m%d")  
204 dates <- dates[seq_len(T)]  
205  
206 y_max <- max(volatilities, na.rm = TRUE)  
207  
208 plot(dates, volatilities[, "HPQ"], type = "l",  
209      col = "red", lwd = 1.5,  
210      ylim = c(0, y_max),  
211      xlab = "Time",  
212      ylab = "Volatility (%)",  
213      main = paste("Multivariate EWMA Volatilities ( $\lambda$  =", round(lambda_est, 4), ")"))  
214  
215 lines(dates, volatilities[, "IBM"], col = "blue", lwd = 1.5)  
216 lines(dates, volatilities[, "SP"], col = "black", lwd = 1.5)  
217  
218 legend("topright",  
219      legend = c("HPQ", "IBM", "S&P Composite"),  
220      col = c("red", "blue", "black"),  
221      lwd = 1.5)  
---
```

Multivariate EWMA Volatilities ($\lambda = 0.9691$)

The figure displays the multivariate EWMA volatility estimates for HPQ, IBM, and the S&P composite index.

The three volatility series exhibit persistent dynamics and co-movement over time, with periods of elevated volatility affecting all assets simultaneously.

Among the three series, HPQ shows the highest volatility levels, followed by IBM, while the S&P composite index displays comparatively lower volatility throughout the sample.

Ex. 10.2

Focus on the monthly log returns of IBM and HPQ stocks from January 1962 to December 2008. Fit a DVEC(1,1) model to the bivariate return series. Is the model adequate? Plot the fitted volatility series and the time-varying correlations.

```
223 #Ex 10.2
224 install.packages("rugarch")
225 install.packages("rmgarch")
226 library(rugarch)
227 library(rmgarch)
228
229 # Log-returns
230 ibm_log <- log(1 + data$IBM)
231 hpq_log <- log(1 + data$HPQ)
232
233 R <- cbind(IBM = ibm_log, HPQ = hpq_log) #creating the matrix
234
235 # Univariate GARCH(1,1) Spec
236 uspec <- ugarchspec(variance.model = list(model = "sGARCH", garchOrder = c(1,1)),
237   mean.model = list(armaOrder = c(0,0), include.mean = TRUE),
238   distribution.model = "norm")
239
240 multispec <- multispec(replicate(2, uspec))
241
242 # DVEC(1,1) Specification
243 dvec.spec <- dccspec(uspec = multispec, dccOrder = c(1,1), model = "DVEC",
244   distribution = "mvnorm")
```

Log-returns for IBM and HPQ are computed and combined into a bivariate return matrix, which serves as input for the multivariate volatility model.

A univariate GARCH(1,1) specification with Gaussian innovations is defined for each series and replicated to construct the multivariate specification.

The DVEC(1,1) model is then specified to allow for time-varying conditional variances and covariances.

```
246 # Model estimation
247 dvec.fit <- dccfit(spec = dvec.spec, data = R, fit.control = list(eval.se = TRUE))
248 dvec.fit
249
250 # STANDARDIZED RESIDUALS (rmgarch: standardize manually)
251 res <- residuals(dvec.fit) # raw residuals: T x 2
252 H <- rcov(dvec.fit) # conditional covariance matrices: 2 x 2 x T
253
254 vol_ibm <- sqrt(H[1,1,])
255 vol_hpq <- sqrt(H[2,2,])
256
257 Z <- cbind(res[,1] / vol_ibm,
258   res[,2] / vol_hpq)
259 colnames(Z) <- c("IBM", "HPQ")
```

A bivariate DCC(1,1) GARCH model with Gaussian innovations is estimated for the IBM and HPQ return series.

The univariate GARCH components exhibit strong volatility persistence for both series, as indicated by large values of the GARCH coefficients.

The estimated DCC parameters suggest time-varying conditional correlations between the two assets.

Standardized residuals are constructed by scaling the model residuals with the corresponding conditional standard deviations extracted from the fitted DVEC covariance matrices.

Ljung–Box tests with 12 lags are then applied to the standardized residuals to assess remaining linear serial correlation, and to the squared standardized residuals to check for remaining ARCH-type dependence.

The test outputs are reported for both IBM and HPQ.

```

*-----*
*           DCC GARCH Fit           *
*-----*

```

```

Distribution      : mvnorm
Model            : DCC(1,1)
No. Parameters    : 11
[VAR GARCH DCC UncQ] : [0+8+2+1]
No. Series       : 2
No. Obs.        : 564
Log-Likelihood   : 1319.942
Av.Log-Likelihood : 2.34

```

Optimal Parameters

```

-----

```

	Estimate	Std. Error	t value	Pr(> t)
[IBM].mu	0.007923	0.002719	2.9139	0.003569
[IBM].omega	0.000384	0.000186	2.0630	0.039115
[IBM].alpha1	0.119747	0.044893	2.6674	0.007645
[IBM].beta1	0.806807	0.064011	12.6042	0.000000
[HPQ].mu	0.011847	0.003922	3.0206	0.002522
[HPQ].omega	0.000599	0.000245	2.4475	0.014387
[HPQ].alpha1	0.067677	0.020867	3.2432	0.001182
[HPQ].beta1	0.866597	0.032244	26.8763	0.000000
[Joint]dccal	0.049355	0.027364	1.8036	0.071292
[Joint]dccbl	0.775098	0.120127	6.4523	0.000000

```

261 # LJUNG-BOX TESTS (residuals and squared residuals)
262 # Serial correlation in residuals
263 lb_ibm <- Box.test(Z[, "IBM"], lag = 12, type = "Ljung-Box")
264 lb_hpq <- Box.test(Z[, "HPQ"], lag = 12, type = "Ljung-Box")
265
266 # Remaining ARCH effects (serial correlation in squared residuals)
267 lb2_ibm <- Box.test(Z[, "IBM"]^2, lag = 12, type = "Ljung-Box")
268 lb2_hpq <- Box.test(Z[, "HPQ"]^2, lag = 12, type = "Ljung-Box")
269

```

Box-Ljung test

```

data: Z[, "IBM"]
X-squared = 8.775, df = 12, p-value = 0.722

```

Box-Ljung test

```

data: Z[, "HPQ"]
X-squared = 5.0585, df = 12, p-value = 0.956

```

Box-Ljung test

```

data: Z[, "IBM"]^2
X-squared = 4.5665, df = 12, p-value = 0.9709

```

Box-Ljung test

```

data: Z[, "HPQ"]^2
X-squared = 12.235, df = 12, p-value = 0.427
data: Z[, "HPQ"]
X-squared = 5.0585, df = 12, p-value = 0.956

```

Box-Ljung test

```

data: Z[, "IBM"]^2
X-squared = 4.5665, df = 12, p-value = 0.9709

```

Box-Ljung test

```

data: Z[, "HPQ"]^2
X-squared = 12.235, df = 12, p-value = 0.427

```

The Ljung–Box tests on the standardized residuals do not indicate remaining serial correlation for either series (IBM: $p = 0.722$; HPQ: $p = 0.956$). Likewise, Ljung–Box tests on the squared standardized residuals do not provide evidence of remaining ARCH effects (IBM: $p = 0.971$; HPQ: $p = 0.427$). Overall, these diagnostics suggest that the DVEC specification adequately captures the main linear dependence and conditional heteroskedasticity patterns in the two return series.

```

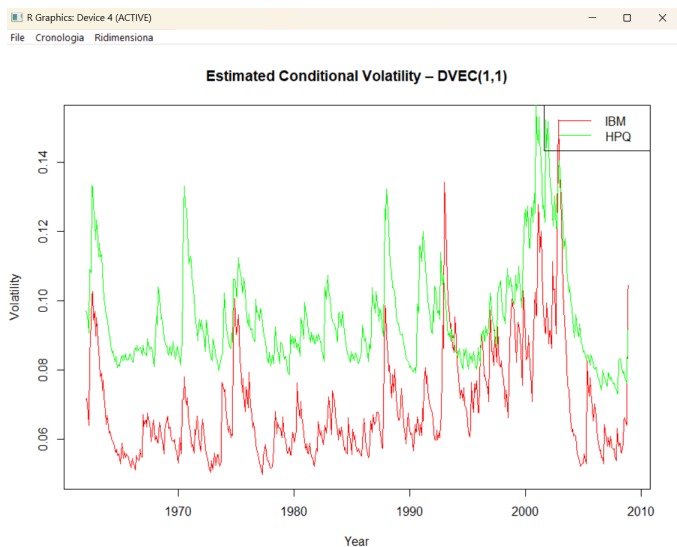
272 # CONDITIONAL VOLATILITIES
273 H <- rcov(dvec.fit)
274 time <- seq(from = 1962, to = 2008 + 11/12, length.out = T)
275 windows()
276
277 plot(time, vol_ibm, type = "l", col = "red", main = "Estimated Conditional Volatility -
278       ylab = "Volatility", xlab = "Year")
279 lines(time, vol_hpq, col = "green")
280 legend("topright", legend = c("IBM", "HPQ"), col = c("red", "green"), lty = 1)
281
282 # TIME-VARYING CORRELATION
283 cov_12 <- H[1,2,]
284 rho_t <- cov_12 / (vol_ibm * vol_hpq)
285
286 plot(time, rho_t, type = "l", col = "darkgreen", main = "Time-Varying Correlation - IBM
287       ylab = "Correlation", xlab = "Year", ylim = c(-0.5, 1))
288 abline(h = mean(rho_t), lty = 2, col = "gray")
289 grid()

```

Conditional volatilities are extracted from the estimated DVEC(1,1) model by taking the square roots of the diagonal elements of the conditional covariance matrices.

A time index matching the sample period is constructed, and the conditional volatility series for IBM and HPQ are plotted jointly to allow a direct comparison of their dynamic behavior.

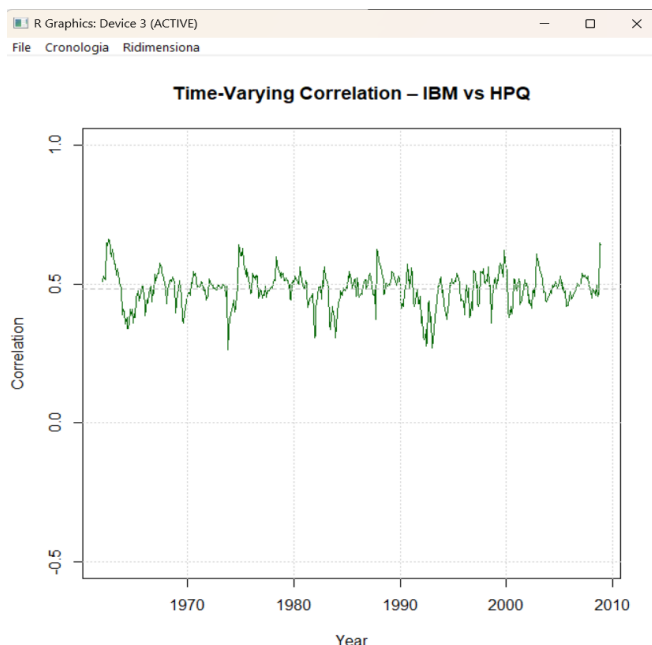
In addition, the time-varying conditional correlation between the two assets is computed from the off-diagonal elements of the covariance matrices and plotted over time, with the sample average correlation reported as a reference.



The figure reports the estimated conditional volatilities of IBM and HPQ obtained from the DVEC(1,1) model.

Both series exhibit persistent volatility dynamics and clear co-movement over time, with HPQ displaying systematically higher volatility levels than IBM.

Periods of increased volatility are shared across the two assets, indicating common sources of risk affecting both return series.



The figure shows the time-varying conditional correlation between IBM and HPQ implied by the DVEC(1,1) model.

The correlation remains positive throughout the sample and fluctuates around its long-run average, as indicated by the horizontal reference line.

These dynamics suggest that the dependence between the two assets is not constant over time but varies in response to changing market conditions.

Ex. 10.3

Focus on the monthly log returns of the S&P composite index and HPQ stock. Build a BEKK model for the bivariate series. What is the fitted model? Plot the fitted volatility series and the time-varying correlations.

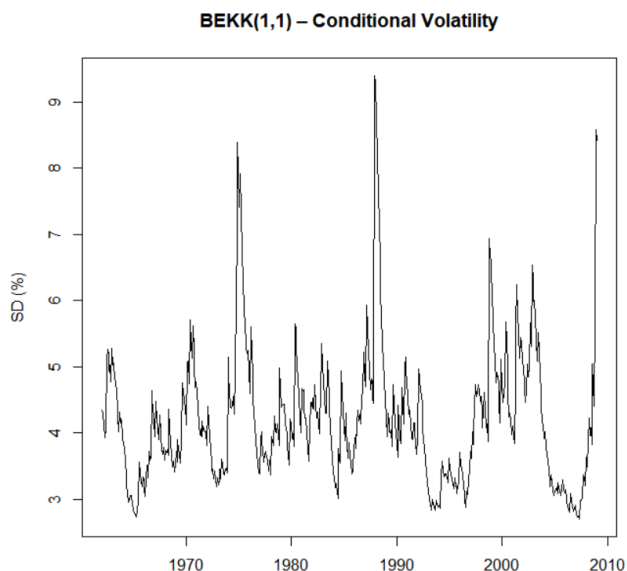
```
303 #Log-returns
304 da <- read.table(dest, header = TRUE)
305 date <- as.Date(as.character(da$date), format = "%Y%m%d")
306
307 rlog <- 100 * log(1 + da[, c("SP", "HPQ")])
308 colnames(rlog) <- c("SP", "HPQ")
309 r_xts <- xts(rlog, order.by = date)
```

As a preliminary step, returns are transformed into log-returns expressed in percentage terms. To model the conditional second moments, a BEKK(1,1) specification is employed. The BEKK model, provides a parsimonious yet flexible parameterization of the conditional covariance matrix. Its main advantage lies in guaranteeing positive definiteness of the covariance matrix by construction, while allowing past shocks and past conditional variances to affect both variances and covariances dynamically. Denoted the vector of returns, the model assumes a constant conditional mean and specifies the conditional covariance matrix as a function of lagged squared innovations and lagged covariance matrices

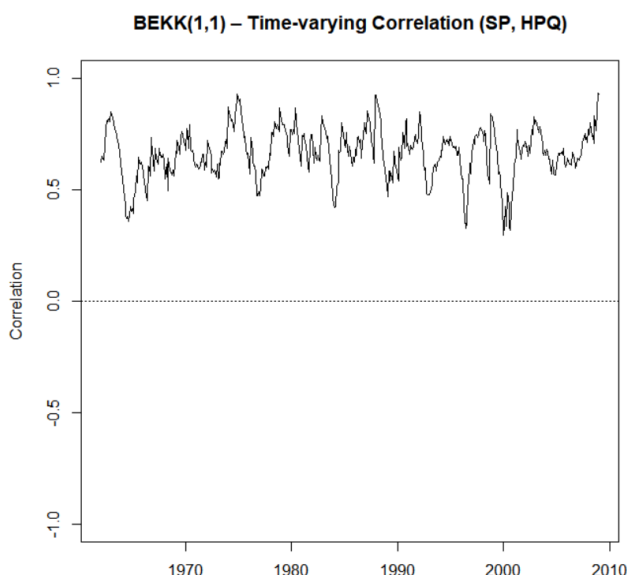
```
311 #BEKK(1,1) Estimation
312 fit <- BEKK11(as.matrix(r_xts), include.mean = TRUE, cond.dist = "normal")
313
314 fit$estimates
315
316 # Conditional volatilities and correlations
317 Sigma <- fit$Sigma.t
318
319 # Conditional standard deviations
320 vol_SP <- sqrt(Sigma[,1])
321 vol_HPQ <- sqrt(Sigma[,4])
322
323 # Conditional correlation
324 rho_t <- Sigma[,2] / sqrt(Sigma[,1] * Sigma[,4])
```

Once the model is estimated, the sequence of conditional covariance matrices is extracted. Conditional volatilities are obtained as the square roots of the diagonal elements of the matrix, while the time-varying conditional correlation is computed by standardizing the conditional covariance. This approach allows for a direct assessment of both individual risk dynamics and the evolving dependence structure between the two assets.

```
326 #Plots
327 windows()
328 plot(date, vol_SP, type = "l",
329       main = "BEKK(1,1) - Conditional Volatility",
330       xlab = "", ylab = "SD (%)")
331 lines(date, vol_HPQ, lty = 2)
332 legend("topright", legend = c("SP", "HPQ"),
333       lty = c(1, 2), bty = "n")
334
335 plot(date, rho_t, type = "l",
336       main = "BEKK(1,1) - Time-varying Correlation (SP, HPQ)",
337       xlab = "", ylab = "Correlation",
338       ylim = c(-1, 1))
339 abline(h = 0, lty = 3)
```



This figure displays the estimated conditional standard deviations of SP and HPQ. Both series exhibit pronounced volatility clustering, characterized by persistent periods of high and low volatility. HPQ is consistently more volatile than the market index, reflecting the higher idiosyncratic risk of an individual stock relative to a diversified portfolio. Several sharp volatility peaks correspond to major episodes of financial stress, including the mid-1970s turmoil, the 1987 stock market crash, the early 2000s downturn, and the global financial crisis of 2008.



This figure reports the time-varying conditional correlation between SP and HPQ returns. The estimated correlation remains positive throughout the sample but exhibits substantial temporal variation. Periods of heightened market volatility are associated with higher correlations, particularly during the late 2000s. This behavior indicates that the co-movement between HPQ and the market strengthens in times of stress, implying a reduction in diversification benefits exactly when risk increases.

Ex. 11.3

Consider the monthly simple excess returns of Pfizer stock and the S&P 500 composite index from January 1990 to December 2003. The excess returns are in `m-pfesp-ex9003.txt` with Pfizer stock returns in the first column.

(a) Fit a fixed-coefficient market model to the Pfizer stock return. Write down the fitted model.

(b) Fit a time-varying CAPM to the Pfizer stock return. What are the estimated standard errors of the innovations to the α_t and β_t series? Obtain time plots of the smoothed estimates of α_t and β_t .

```
342 # Ex. 11.3 - Fixed Market Model + Time-Varying CAPM
343 install.packages("dlm")
344 library(dlm)
345
346 dat <- read.table("m-pfesp-ex9003.txt", header = FALSE)
347 colnames(dat) <- c("PFE", "MKT")
348
```

We begin the analysis by estimating the standard fixed-coefficient market model for Pfizer's monthly excess returns. This specification corresponds to the static version of the CAPM and serves as a benchmark against which the time-varying model in part (b) will be compared.

We estimate the standard market model by ordinary least squares (OLS):

```
354 mm_ols <- lm(PFE ~ MKT, data = dat)
355
356 print(summary(mm_ols))
357
358 # Fitted coefficients
359 alpha_hat <- coef(mm_ols)[1]
360 beta_hat <- coef(mm_ols)[2]
361
362 cat("\nFITTED MODEL (OLS):\n")
363 cat(sprintf("r_PFE,t = %.6f + %.6f * r_MKT,t + e_t\n", alpha_hat, beta_hat))
```

```
Call:
lm(formula = PFE ~ MKT, data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-0.215721 -0.042147 -0.000223  0.042383  0.197424

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.012181   0.005151   2.365   0.0192 *
MKT          0.851257   0.118780   7.167  2.4e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.06646 on 166 degrees of freedom
Multiple R-squared:  0.2363,    Adjusted R-squared:  0.2317
F-statistic: 51.36 on 1 and 166 DF,  p-value: 2.396e-11

FITTED MODEL (OLS):
Browse[1]> cat(sprintf("r_PFE,t = %.6f + %.6f * r_MKT,t + e_t\n", alpha_hat, beta_hat))
r_PFE,t = 0.012181 + 0.851257 * r_MKT,t + e_t
Browse[1]> |
```

The estimated slope coefficient is $\beta = 0.851$ ($p < 0.001$), indicating that Pfizer's excess returns co-move positively with the market factor. Since β is below 1, Pfizer exhibits, on average, lower systematic risk than the market portfolio over the sample. The intercept estimate ($\alpha = 0.012$) is statistically significant at the 5% level ($p = 0.019$), which can be read as evidence of a positive Jensen's alpha within a one-factor CAPM. Economically, however, this interpretation should be treated with caution: alpha can absorb omitted risk factors and time variation in exposures. The market model explains a moderate share of the total variation ($R^2 = 0.236$), consistent with substantial idiosyncratic risk at the individual-stock level.

```

365 #Time-varying CAPM
366
367 build_tvcapm <- function(parm) {
368
369   # Log-parameterization -> positivity
370   V <- exp(parm[1]) # observation variance (eps_t)
371   w1 <- exp(parm[2]) # state innovation variance for alpha_t
372   w2 <- exp(parm[3]) # state innovation variance for beta_t
373
374   # dlmModReg creates a regression DLM with time-varying regressor.
375   # addInt=TRUE => state dimension = 2 (alpha, beta).
376   # dV is scalar; dW must be a vector of length 2 (NOT a matrix).
377   dlmModReg(X = dat$MKT, addInt = TRUE, dV = V, dW = c(w1, w2))
378 }
379
380 #MLE estimation
381 init_parm <- log(c(var(resid(mm_ols)), 1e-6, 1e-6))
382 fit_mle <- dlmMLE(
383   y      = dat$PFE,
384   parm   = init_parm,
385   build  = build_tvcapm,
386   control = list(maxit = 500)
387 )

```

In this part, a time-varying CAPM is estimated using a state-space representation and maximum likelihood via the Kalman filter. The objective is to allow both the intercept (alpha) and the market beta to evolve dynamically over time, rather than being fixed as in the standard CAPM.

The model is implemented using the `dlmModReg` function, which correctly handles time-varying regressors within a dynamic linear model framework. To ensure positivity of the variance parameters, all variances are parameterized in logarithmic form and estimated by maximum likelihood.

Maximum likelihood estimation is carried out by minimizing the negative log-likelihood of the Kalman filter. The reported value corresponds to the log-likelihood evaluated at the estimated parameters and provides a global measure of in-sample fit.

```

Browse[1]> cat("MLE log-likelihood:", -fit_mle$value, "\n\n")
MLE log-likelihood: 352.3407

```

```

389 cat("MLE log-likelihood:", -fit_mle$value, "\n\n")
390
391 # Rebuild model at MLE parameters
392 tvcapm <- build_tvcapm(fit_mle$par)
393
394 #parameters
395 cat("Estimated variance parameters (MLE):\n")
396 cat(sprintf("V (measurement variance eps_t)           = %.10e\n", tvcapm$V))
397 cat(sprintf("w_alpha (innov var eta_alpha,t)         = %.10e\n", tvcapm$W[1, 1]))
398 cat(sprintf("w_beta (innov var eta_beta,t)           = %.10e\n", tvcapm$W[2, 2]))
399
400 sd_alpha_innov <- sqrt(tvcapm$W[1, 1])
401 sd_beta_innov  <- sqrt(tvcapm$W[2, 2])
402
403 cat("\nInnovation STANDARD DEVIATIONS (asked in the exercise):\n")
404 cat(sprintf("sd(eta_alpha,t) = %.10e\n", sd_alpha_innov))
405 cat(sprintf("sd(eta_beta,t)  = %.10e\n", sd_beta_innov))

```

Innovation STANDARD DEVIATIONS (asked in the exercise):

```
Browse[1]> cat(sprintf("sd(eta_alpha,t) = %.10e\n", sd_alpha_innov))
sd(eta_alpha,t) = 2.5670001234e-30
Browse[1]> cat(sprintf("sd(eta_beta,t) = %.10e\n", sd_beta_innov))
sd(eta_beta,t) = 6.8238705687e-02
```

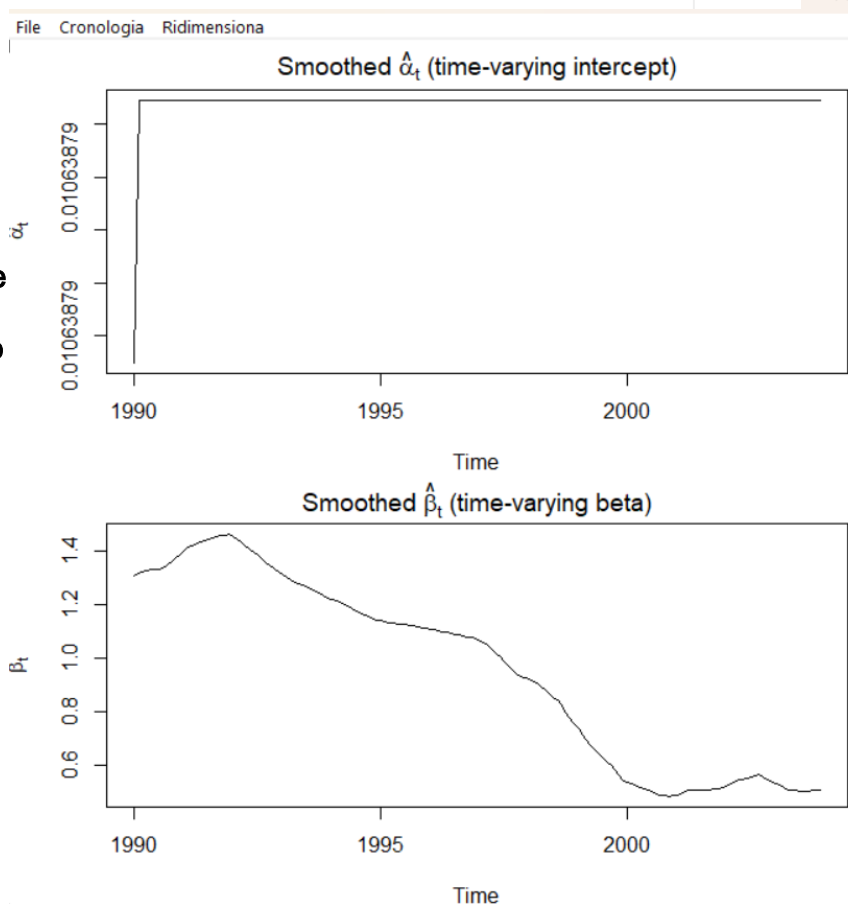
The innovation variance of the intercept equation is essentially zero, implying that the intercept is statistically time-invariant over the sample period. In other words, there is no evidence of a time-varying abnormal return component once market excess returns are taken into account.

In contrast, the innovation variance associated with the market beta is strictly positive and economically meaningful. This indicates that the market exposure of Pfizer varies over time, and that allowing Beta to evolve dynamically improves the model's ability to capture changes in systematic risk.

```
407 #Kalman filter + smoother
408
409 filt <- dlmFilter(dat$PFE, tvcapm)
410 smo <- dlmSmoother(filt)
411
412 alpha_sm <- drop(smo$s[-1, 1])
413 beta_sm <- drop(smo$s[-1, 2])
414
415 #Plots of smoothed alpha_t and beta_t
416 windows()
417 oldpar <- par(no.readonly = TRUE)
418 par(mfrow = c(2, 1), mar = c(4, 4, 2, 1))
419
420 Tn <- length(alpha_sm)
421 time_index <- seq(from = as.Date("1990-01-01"), by = "month", length.out = Tn)
422
423 plot(time_index, alpha_sm, type = "l",
424       xlab = "Time", ylab = expression(hat(alpha)[t]),
425       main = expression("Smoothed " * hat(alpha)[t] * " (time-varying intercept)"))
426
427 plot(time_index, beta_sm, type = "l",
428       xlab = "Time", ylab = expression(hat(beta)[t]),
429       main = expression("Smoothed " * hat(beta)[t] * " (time-varying beta)"))
430
431 par(oldpar)
```

The upper panel shows that the smoothed intercept is essentially flat over the entire sample period. After a short initial adjustment phase, alpha converges to a constant level and exhibits virtually no time variation. This behavior is consistent with the estimated innovation variance of the intercept equation being numerically close to zero, indicating that there is no statistically significant evidence of a time-varying abnormal return component for Pfizer once market excess returns are taken into account.

In contrast, the lower panel highlights substantial time variation in the smoothed market beta. At the beginning of the sample, Pfizer displays a relatively high market exposure, with a beta above one.



Over time, Beta follows a clear downward trend, particularly during the second half of the 1990s, and reaches values well below one around the early 2000s. This pattern suggests a progressive reduction in Pfizer's sensitivity to systematic market risk, potentially reflecting changes in the firm's business mix, risk profile, or the defensive nature of the pharmaceutical sector during market downturns.

Overall, the smoothed state estimates indicate that the dynamic behavior of Pfizer's excess returns is driven primarily by a time-varying market beta, while the intercept remains stable. These results support the use of a time-varying CAPM and show that allowing for a flexible beta captures economically meaningful changes in systematic risk over time.